Nonlinear Relational Invariants: Joint Generation, Quotient Obstruction, and Refinement Geometry

Draft section for a later version of *Observable Equivalence and Relational Invariants in Recursive Generative Systems* (RGG-2.1, not merged)

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Status labels. **Proved**: an argument is given here. **Checked**: finite computation only. **Proposed**: a definition or choice. **Open**: not settled. The operation $\vee$ below is a *proposed* joint-generation rule; nothing here asserts that earlier UNNS grammar defines it, and no physical claim is made.

1 Setting

By Corollary 10.2, an integer nest output $M \perp N$ has $N \mid M$; writing $M = kN$, $d = N + 1 \geq 2$ gives value $v(r) = kd^2$ for the *anchor route* $r = (k, d)$. For $m \geq 1$ let $F(m) = \{r : v(r) = m\}$ (the *fibre*); $|F(m)| = R(m)$. Field occurrences ($N \nmid M$) are outside this section.

Definition 1.1 (Joint continuation $\vee$; proposed)**.** $r \vee s = (\mathrm{lcm}(k_r, k_s),\ \mathrm{lcm}(d_r, d_s))$. Route order: $r \leq s$ iff $k_r \mid k_s$ and $d_r \mid d_s$.

Lemma 1.2 (Order structure; proved)**.** *$\vee$ is the least upper bound for $\leq$ (hence commutative, associative, idempotent); $v(r) \mid v(r \vee s)$ and $v(s) \mid v(r \vee s)$; v is strictly increasing in each coordinate, so every fibre is an antichain.*

Proof. Lub: componentwise lcm. Divisibility of values is immediate from kd^2. If $r \leq r'$ with $v(r) = v(r')$ then $k'/k \cdot (d'/d)^2 = 1$ with both ratios positive integers, so $r = r'$. □

2 Fibre separation and relational distance

Theorem 2.1 (Join value; proved)**.** *For $r = (m/d^2, d)$, $r' = (m/d'^2, d') \in F(m)$ put $g = \gcd(d, d')$, $L = \mathrm{lcm}(d, d')$. Then $\mathrm{lcm}(k, k') = m/g^2$ and*

$$v(r \vee r') = m\Big(\frac{L}{g}\Big)^2.$$

Proof. For a prime p with $\nu_p(m) = e$, $\nu_p(d) = a$, $\nu_p(d') = a'$: $\nu_p \,\mathrm{lcm}(k, k') = \max(e - 2a, e - 2a') = e - 2\min(a, a') = \nu_p(m/g^2)$. Multiply by L^2. □

Corollary 2.2 (Universal pairwise separation; proved)**.** *If $r \neq r'$ in $F(m)$ then $v(r \vee r') > m = v(r \vee r)$. Hence the joint-continuation value with probe $s = r$ distinguishes r from every other route of its fibre. (This is a consequence of the definition of $\vee$, not independent evidence of interaction.)*

Proof. $r \neq r'$ forces $d \neq d'$ (as $k = m/d^2$), so $L > g$. □

Definition 2.3 (Relational distance). $D(r,r') = \log\big(v(r\vee r')/m\big) = 2\log(L/g) = 2\sum_p |\nu_p(d) - \nu_p(d')|\log p$ for $r, r' \in F(m)$.

Theorem 2.4 (Fibre geometry; proved). *D is a metric on $F(m)$. With $b_p = \lfloor \nu_p(m)/2 \rfloor$, $F(m)$ is isometric to $\prod_p \{0,\dots,b_p\} \setminus \{0\}$ with weighted ℓ^1 metric (weights $2\log p$); so $|F(m)| = \prod_p (b_p+1) - 1 = R(m)$. Its diameter is $2\sum_p b_p \log p$ if at least two $b_p \geq 1$, and $2(b_p-1)\log p$ if exactly one.*

Proof. $d \mapsto (\nu_p(d))_p$ identifies $F(m)$ with exponent vectors $0 \leq a_p \leq b_p$ other than 0 (the condition $d^2 \mid m$); the formula for D is the weighted ℓ^1 distance. Diameter: the full box has diameter $2\sum b_p \log p$ attained by opposite corners; if at least two primes have $b_p \geq 1$, split them into two nonempty groups: the point equal to b on the first group and 0 elsewhere, and the point equal to b on the second group and 0 elsewhere, are both nonzero and at full distance; if only one, the origin removal shortens the chain by one step. □

3 Refinement geometry

For a prime q put $T_q(k,d) = (k/q^2, dq)$ when $q^2 \mid k$. It preserves v; its inverse is $(k,d) \mapsto (kq^2, d/q)$ when $q \mid d$ and $d/q \geq 2$.

Theorem 3.1 (Joint covariance; proved). *If $T_q r, T_q s$ are defined, $T_q r \vee T_q s = T_q(r \vee s)$. Consequently v of the joint continuation, and D, are invariant under joint refinement, while one refinement step moves a route by $D(T_q r, r) = 2\log q$. Invariance under refinement of* one *argument fails: $v(T_q r \vee s) \neq v(r \vee s)$ in* 5098 *of* 10304 *tested cases (checked).*

Proof. $\operatorname{lcm}(k/q^2, k'/q^2) = \operatorname{lcm}(k,k')/q^2$ and $\operatorname{lcm}(dq, d'q) = q\operatorname{lcm}(d,d')$. □

Theorem 3.2 (Distance equals refinement cost; proved). *Give each refinement step on prime q the cost $2\log q$. Then for $r, r' \in F(m)$ the minimal total cost of a chain of refinement steps (and inverses) from r to r' equals $D(r,r')$.*

Proof. A step changes one coordinate a_p by ± 1, so cost $\geq \sum_p |a_p - a'_p|\, 2\log p = D$. Conversely, the vector $\max(a,a')$ is $\leq b$ coordinatewise and nonzero, hence a fibre route (it is $d = \operatorname{lcm}(d,d')$); the path $a \to \max(a,a') \to a'$ stays in the fibre (never at $d=1$) and costs exactly D. □

4 A set-theoretic quotient criterion (no bilinearity)

Let Π be an observation, $\sim_\Pi$ the induced equivalence, and J a (possibly set-valued) joint operation; $\Pi_* J(r,s)$ denotes the image set. Say J *descends* to Π if $r \sim_\Pi r' \Rightarrow \Pi_* J(r,s) = \Pi_* J(r',s)$ and likewise in the second slot.

Theorem 4.1 (Set-theoretic Quotient Criterion; proved). *Let $\mathcal{T}$ be a family of refinements and $\sim_\mathcal{T}$ the equivalence generated by $\{(r, Tr)\}$. Assume $\sim_\mathcal{T} = \sim_\Pi$. Then J descends to Π iff $\Pi_* J(Tr,s) = \Pi_* J(r,s)$ and $\Pi_* J(r,Ts) = \Pi_* J(r,s)$ for all generating T, r, s.*

Proof. For fixed s the relation $\{(r,r') : \Pi_* J(r,s) = \Pi_* J(r',s)\}$ is an equivalence relation; it contains $\sim_\mathcal{T}$ iff it contains the generators. Same for the second slot. □

Corollary 4.2 (Application; proved/checked). *Hypothesis $\sim_\mathcal{T} = \sim_{\text{value}}$ holds for anchor routes (Theorem 3.2: fibres are connected by prime refinements). The product $J_\times(r,s) = (k_r k_s, d_r d_s)$ satisfies value-invariance and descends (checked); $\vee$ violates one-argument invariance and does not descend.*

Definition 4.3 (Set deficit). Let $\kappa(m)$ be the number of classes of the largest J-congruence inside $\sim_\Pi$ on $F(m)$, and $\delta_{\text{set}}(m) = \kappa(m) - 1$. Then $\delta_{\text{set}}(m) = R(m) - 1$ for $\vee$ (Corollary 2.2) and 0 for $J_\times$ with value observation.

5 Single-probe fibre reconstruction

Let $h(m) = \prod_p p^{\lfloor \nu_p(m)/2 \rfloor}$ and $s_m = (m/h(m)^2,\ h(m))$.

Theorem 5.1 (Single-Probe Fibre Reconstruction; proved)**.** *$F(m) \neq \varnothing$ iff m is not squarefree iff $h(m) \geq 2$, and then $s_m \in F(m)$. For every $r = (m/d^2, d) \in F(m)$,*

$$r \vee s_m = \big(m/d^2,\ h(m)\big), \qquad v(r \vee s_m) = m\Big(\tfrac{h(m)}{d}\Big)^2.$$

Hence $d = h(m)\sqrt{m/v(r \vee s_m)}$ and the entire route r is recovered from one joint-continuation value with the fixed probe s_m.

Proof. $d^2 \mid m$ implies $\nu_p(d) \leq \lfloor \nu_p(m)/2 \rfloor$ for every p, so $d \mid h$; in particular $h^2 \mid m$ and $s_m \in F(m)$ when $h \geq 2$. Then $\operatorname{lcm}(d, h) = h$ and $m/h^2 \mid m/d^2$ gives $\operatorname{lcm}(m/d^2, m/h^2) = m/d^2$. The value is $(m/d^2)h^2$. □

Corollary 5.2 (Probe family; proved)**.** *Any probe $s = (k_s, d_s)$ with $h(m) \mid d_s$ and $k_s \mid m/h(m)^2$ resolves $F(m)$: $\operatorname{lcm}(d, d_s) = d_s$ for all $d \mid h$, and $\operatorname{lcm}(m/d^2, k_s) = m/d^2$, so $v(r \vee s) = (m/d^2)d_s^2$ is injective in d. (Sufficient, not claimed necessary.) Probes need not lie in the fibre.*

Remark 5.3. s_m depends on m, so the probe is fixed per fibre, not universal across fibres. Checked: all 7840 nonempty fibres with $m \leq 20000$; fibre of 72: $h = 6$, $s_{72} = (2, 6)$, join values $648, 288, 72$ for $(18, 2), (8, 3), (2, 6)$.

6 Robustness under coordinate choice, and an erratum

Lemma 6.1 (Separation is coordinate-robust; proved)**.** *Let routes be coordinates in a poset chart with a value v strictly increasing in each coordinate, and let J satisfy $J(r, s) \geq r, s$ coordinatewise. Then for distinct r, r' with equal value, $v(J(r, r')) > v(r)$.*

Proof. If $J(r, r') = r$ then $r' \leq r$; with equal value strict monotonicity forces $r' = r$. So $J(r, r') \geq r, \neq r$, and strict monotonicity gives $v(J(r, r')) > v(r)$. □

This applies to joins in (k, d), in (k, N), and in occurrence coordinates (M, N) (value $M(N+1)^2/N$ is strictly increasing in M and in $N \geq 1$; the occurrence join is anchor-closed because $\operatorname{lcm}(N, N') \mid \operatorname{lcm}(M, M')$). Checked on 2745 same-value pairs ($m \leq 6000$). The exact formulas of Theorems 2.1–3.2 are specific to (k, d) coordinates; that s_m also resolves every fibre under the other two joins is checked (693/693), not proved.

Erratum (RGG-2 note). The statement "the mixed operations stay in the fibre, so the fibre is a lattice" is false on the anchor domain $d \geq 2$. $(\operatorname{lcm} k, \gcd d)$ returns $(m/g^2, g)$, which is inadmissible when $g = \gcd(d, d') = 1$: e.g. $(18, 2), (8, 3) \mapsto (72, 1)$ (501 of 2745 same-value pairs). $(\gcd k, \operatorname{lcm} d) = (m/L^2, L)$ is always admissible and is the least upper bound in the d-divisibility order. Hence $F(m)$ is a *join-semilattice*, not a lattice (pairs with $\gcd(d, d') = 1$ have no common lower bound); it becomes a lattice when the formal route $(m, 1)$ is adjoined (checked, as it is the full grid $\prod\{0..b_p\}$).

7 Summary of status

Statement	Status
Order structure of $\vee$; fibres are antichains	proved
$v(r \vee r') = m(L/g)^2$; universal pairwise separation	proved
D is a weighted ℓ^1 metric; fibre = box minus origin	proved
Joint covariance; D invariant; one-argument invariance fails	proved / checked
Distance = minimal refinement cost	proved
Set-theoretic quotient criterion	proved
Single-probe reconstruction theorem; probe family	proved
Coordinate-robust separation (lemma)	proved
Same single probe under $(k, N), (M, N)$ joins	checked only
Excess $\varepsilon = v(r \vee s)/\operatorname{lcm}(v(r), v(s))$ not a function of values	checked (e.g. values $(4, 36)$ give 1 and 4)
Selection of $\vee$ by an independent generative constraint	open
Bilinear representation of $\vee$; field occurrences; physical meaning	open / not claimed

Verifiers: `rgg2.py` (23 checks, 5 data rows) and `rgg21.py` (14 checks, 2 data rows), deterministic, exact integer arithmetic.